PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Automated mechanical cardiopulmonary resuscitation devices versus manual chest compressions in the treatment of cardiac arrest: Protocol of a systematic review and meta-analysis comparing machine to human
AUTHORS	Obermaier, Manuel; Zimmermann, Johannes; Popp, Erik; Weigand, Markus; Weiterer, Sebastian; Dinse-Lambracht, Alexander; Muth, Claus-Martin; Nußbaum, Benedikt; Grasner, Jan-Thorsten; Seewald, Stephan; Jensen, Katrin; Seide, Svenja

VERSION 1 – REVIEW

REVIEWER	Ryan Brandon Hunter, MD and Ryan W. Morgan, MD, MTR Children's Hospital of Philadelphia Perelman School of Medicine at the University of Pennsylvania Philadelphia, Pennsylvania, United States of America
REVIEW RETURNED	29-Aug-2020

GENERAL COMMENTS	Summary and Major Comments: This manuscript presents a protocol for a planned systematic review of mechanical versus manual CPR in adult patients. The authors note that to date there have been conflicting data regarding outcomes, and that consensus statements favor manual CPR as the standard of care, allowing for consideration of mechanical CPR in specific circumstances (e.g. limited rescuers present, prolonged CPR, etc.). That said, use of mechanical CPR devices has increased over recent years (1.9% to 8% from 2010-2016, Kahn et al 2016). The authors acknowledge a primary knowledge gap left by prior systematic reviews to be addressed as follows: 1) it is unknown how semi-circumferential band devices perform relative to piston devices; 2) it is unknown whether certain subpopulations may benefit from mechanical CPR; and 3) it is unclear if the timing of mechanical CPR device application may show benefit. The proposed review would broadly search Medline and Embase databases to assess primary outcomes of survival with good neurologic function and ROSC, as well as secondary outcomes of short- and long-term survival, resuscitation trauma, and CPR quality surrogates. They intend subgroup analyses of piston vs band devices and in-hospital vs out-of-hospital cardiac arrest groups. Overall, the evaluation of the knowledge gap and conflicting evidence is sound. Our greatest concern is whether there is enough data currently published to adequately address these issues. Additionally, there is, as noted below, some discordance between the identified weaknesses of previous reviews and the objectives and methods of the present review. The authors note a recent (2018) Cochrane review that failed to address piston-

	versus band-based devices and subpopulation differences. It should be noted that the Cochrane reviewers did intend a priori subgroup analyses of device times as well as IHCA versus OHCA, but abandoned pooled subgroup analysis because of insufficient and variably collected data. The authors note that the largest three RCTs provide the richest sources of patient data, and each of them (PARAMEDIC, LINC, and CIRC) were included in the Cochrane review. As the authors move forward with the review, it would be important to highlight their subgroup analyses of IHCA versus OHCA, piston versus band, and evaluation of timing of device application, as these features will most add to the prior reviews that have been performed Specific Comments: - In general, the Introduction is quite long. While it provides and accurate broad view, we would advise cutting this down and more succinctly get to the knowledge gap being evaluated. - Line 107 – The data review in the first two paragraphs is likely unnecessary; most of the relevant information is shared under "Burden of disease." We would recommend condensing the first two paragraphs to 1-2 sentences and place in the beginning of "Burden of disease." - Lines 111-112 – please specify in- versus out-of-hospital cardiac arrest - 121-128 – Consider condensing this paragraph to more concisely make the point – e.g. "Many young and able-bodied patients who suffer sudden cardiac arrest have the potential for recovery with good quality of life, but this recovery hinges on high quality CPR at the time of the event." - Lines 146-148 – Please include specific citation for this statement. - Lines 206-207 – The knowledge gaps defined earlier specified that it is: 1) unknown if piston vs band CPR devices perform better, and 2) whether there are subpopulations or timing of mechanical CPR device application may show benefit. Addressing these in the objectives would be important. - Lines 208-209 – I would recommend placing the definition of cardiac arrest in Methods section below where other terms are defined [of note, this term is defined again in line 364] - Line 253 – Given the knowledge gaps described, will there be an effort to address comparison of mechanical devices, timing of application, or identification of subgroups that may benefit? - Line 278 – The authors should cite how they created the list of available devices (to ensure it is exhaustive). - The authors discuss limitations of previous systematic reviews, but it is not entirely clear how this review will differ. Please utilize the Discussion section to differentiate the planned study from previous work and elaborate on the specific gap that this will fill that the 2018 Cochrane review failed to address. Please also further elaborate on potential limitations of the planned study. Grammatical: - Line 65 – "reviewers" is spelled incorrectly. - Line 126 – incident, not incidence - Line 201 - delay not dalay - Line 218 – "valid recommendations" - Line 483 – dependent not dependant

REVIEWER	Keith Couper University of Warwick, UK I have previously published work on mechanical chest compression devices (including systematic reviews).
REVIEW RETURNED	08-Sep-2020

GENERAL COMMENTS	Thank you for submitting this systematic review protocol paper to BMJ Open. Introduction The introduction is rather odd. The authors appear to be drawing parallels between mechanical chest compression devices and rectal application of tobacco smoke. The authors question whether mechanical CPR devices should be included in Advanced Life Support algorithms. Mechanical chest compression devices are already included in Advanced Life Support guidelines, and their use is supported in some (albeit very specific) circumstances. It is misleading to suggest that all use should be abandoned. The key clinical question (which is mentioned in the second to last sentence) is whether devices should be routinely used. The need for this review is justified on the basis on perceived limitations of previous systematic reviews. It is unclear when the authors undertook searches to identify previous reviews, but it appears that key recent reviews (and previous reviews) have not been identified. As such, it is unclear why a further review is required. Outcomes How will you address the issue of ROSC being recorded at different time points in studies? (e.g. any ROSC, ROSC>20 minutes, ROSC at hospital handover). Where multiple time points are used, which will you use? Why is neurological outcome not considered after discharge/ 30-days? Please review your outcome list and check consistency with PROSPERO registration. Eligibility criteria The introduction notes the publication of three large randomised controlled trials. What is the rationale for including quasi-RCTs? General comments It would be informative for the reader to provide more information about the timeline of this project. The PROSPERO record is recorded as being completed in 2017, with a start date of 2016. . The record also states: “An abstract, and oral presentation containing partial aspects of a preliminary analysis have been publicized in German language at the scientific meeting “Deutscher Anästhesie Congress” in Düsseldorf, Germany, on May 7th, 2015.” Search strategy

	It would be helpful to provide information on how you developed your search strategy. Was an information scientist involved? Did you consider the inclusion of MESH terms (e.g. “cardiopulmonary resuscitation” with the instrumentation qualifier)? Public and patient involvement It would be more appropriate to simply detail that patients and the public were not involved. Many studies of cardiac arrest have successfully included patients and the public in their design and oversight (e.g. PARAMEDIC trial). General readability I would recommend careful proof-reading of the paper.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Summary and Major Comments: This manuscript presents a protocol for a planned systematic review of mechanical versus manual CPR in adult patients. The authors note that to date there have been conflicting data regarding outcomes, and that consensus statements favor manual CPR as the standard of care, allowing for consideration of mechanical CPR in specific circumstances (e.g. limited rescuers present, prolonged CPR, etc.). That said, use of mechanical CPR devices has increased over recent years (1.9% to 8% from 2010-2016, Kahn et al 2016). The authors acknowledge a primary knowledge gap left by prior systematic reviews to be addressed as follows: 1) it is unknown how semi-circumferential band devices perform relative to piston devices; 2) it is unknown whether certain subpopulations may benefit from mechanical CPR; and 3) it is unclear if the timing of mechanical CPR device application may show benefit. The proposed review would broadly search Medline and Embase databases to assess primary outcomes of survival with good neurologic function and ROSC, as well as secondary outcomes of short- and long-term survival, resuscitation trauma, and CPR quality surrogates. They intend subgroup analyses of piston vs band devices and in-hospital vs out-of-hospital cardiac arrest groups.

Overall, the evaluation of the knowledge gap and conflicting evidence is sound. Our greatest concern is whether there is enough data currently published to adequately address these issues. Additionally, there is, as noted below, some discordance between the identified weaknesses of previous reviews and the objectives and methods of the present review. The authors note a recent (2018) Cochrane review that failed to address piston- versus band-based devices and subpopulation differences. It

should be noted that the Cochrane reviewers did intend a priori subgroup analyses of device times as well as IHCA versus OHCA, but abandoned pooled subgroup analysis because of insufficient and variably collected data. The authors note that the largest three RCTs provide the richest sources of patient data, and each of them (PARAMEDIC, LINC, and CIRC) were included in the Cochrane review. As the authors move forward with the review, it would be important to highlight their subgroup analyses of IHCA versus OHCA, piston versus band, and evaluation of timing of device application, as these features will most add to the prior reviews that have been performed

Thank you for your encouraging comment on our proposed protocol. Numerous trials – and as reviewer #2 correctly stated, several reviews – have been published trying to assess which chest compression technique is “better” – but as you point out, the question should be which patients (in which siutations, und under which circumstances, and when) do profit from eiter one or the other thechniqe. Therefore, the rationale for this review – and why the authors are convinced that it might provide further understanding – is to adress this issue by its subruop analyses. Beside piston vs. band and in- vs. out-of-hospital cardiac arrest, initially presenting rhythm (shockable, non-shockable), and timing, we have also to consider regional (urban vs. rural, European vs. American systems) variabilities in health care. As once again made clear in the current pandemic situation, that data from one health care system may not be applied to another health care system without considering specific structural differences. In this case, some European countries provide a two-tiered EMS system consisting of physician-staffed advanced care vehicles and paramedic-staffed ambulances, while for example the American system is based on a paramedic system.

Specific Comments:

- In general, the Introduction is quite long. While it provides and accurate broad view, we would advise cutting this down and more succinctly get to the knowledge gap being evaluated.

Following the comments of both reviewers, we revised and shortened our complete introduction section.

- Line 107 – The data review in the first two paragraphs is likely unnecessary; most of the relevant information is shared under “Burden of disease.” We would recommend condensing the first two

paragraphs to 1-2 sentences and place in the beginning of “Burden of disease.”
We adhered to your recommendation, condenesing and switching the two paragraphs.

- Lines 111-112 – please specify in- versus out-of-hospital cardiac arrest

Following your recommendation, we specified in- and out-of-hospital settings in the section “Population characteristics and setting”.

- 121-128 – Consider condensing this paragraph to more concisely make the point – e.g. “Many young and able-bodied patients who suffer sudden cardiac arrest have the potential for recovery with good quality of life, but this recovery hinges on high quality CPR at the time of the event.”

We adopted your suggestion and placed this paragraoph under “rationale”.

- Lines 146-148 – Please include specific citation for this statement.
- Lines 206-207 – The knowledge gaps defined earlier specified that it is: 1) unknown if piston vs band CPR devices perform better, and 2) whether there are subpopulations or timing of mechanical CPR device application may show benefit. Addressing these in the objectives would be important.
- Lines 208-209 – I would recommend placing the definition of cardiac arrest in Methods section below where other terms are defined [of note, this term is defined again in line 364]
- Line 253 – Given the knowledge gaps described, will there be an effort to address comparison of mechanical devices, timing of application, or identification of subgroups that may benefit?
- Line 278 – The authors should cite how they created the list of available devices (to ensure it is exhaustive).

We appreciate your helpful indications and updated the particular sections in order to clarify any questions and to implement your advices.

- The authors discuss limitations of previous systematic reviews, but it is not entirely clear how this review will differ. Please utilize the Discussion section to differentiate the planned study from previous

work and elaborate on the specific gap that this will fill that the 2018 Cochrane review failed to address. Please also further elaborate on potential limitations of the planned study.
We appreciate your recommendation and extended the proposed items in our discussion section. Additional justification we give in our replies to the introductory comments of reviwer #1 (above), and of reviewer #2 (below), respectively.

Grammatical:

- Line 65 – “reviewers” is spelled incorrectly.
- Line 126 – incident, not incidence
- Line 201 - delay not dalay

Thank you for your valuable hints, we corrected the typos.

- Line 218 – “valid recommendations”

We hope that we well understood your suggestion and adjusted the sentence.

- Line 483 – dependent not dependant

As the journal is based in the UK, we anticipated that the journal might prefer the British English spelling with the letter ”a”.

Reviewer: 2

Thank you for submitting this systematic review protocol paper to BMJ Open.

Introduction

The introduction is rather odd. The authors appear to be drawing parallels between mechanical chest compression devices and rectal application of tobacco smoke. The authors question whether mechanical CPR devices should be included in Advanced Life Support algorithms.

By introducing the “milestones” of resuscitation and contrast with some – as you stated – odd (and therefore discontinued) ideas the authors were trying to achieve a kind of humoristic approach to the

topic. We deleted this section, as it also unnecessarily prolonged our – as reviewer #1 correctly pointed out – very long introduction.

Mechanical chest compression devices are already included in Advanced Life Support guidelines, and their use is supported in some (albeit very specific) circumstances. It is misleading to suggest that all use should be abandoned. The key clinical question (which is mentioned in the second to last sentence) is whether devices should be routinely used.
The need for this review is justified on the basis on perceived limitations of previous systematic reviews. It is unclear when the authors undertook searches to identify previous reviews, but it appears that key recent reviews (and previous reviews) have not been identified. As such, it is unclear why a further review is required.

The authors entirely agree with this statement. Of course, mechanical CPR is introduced in the ALS guidelines, so we precised this paragraph as we intended to put its “routine” use into question – as reviewer #2 accurately pointed out. We have taken note of your remarks and included some newer reviews to the “need for the systematic review” and “discussion” sections. As previously mentioned in our reply to reviewer #1 – who also underlined the importance of subgroup analysis – we hope our review may contribute to change the focus of future trials – from a “traditional” comparison of survival rates in order to recommend “routine use” towards a strategy trying to identify specified groups of risk or benefit. We consider this as an extremely relevant and current clinical issue with regard to increasing individualized treatment options in hospital, when mechanical CPR might be used as a “bridge” to definitive treatment for designated patient groups under certain conditions and in specific environments. With ongoing registration and publication of trials on mechanical CPR issues, we may additionally provide an update of existing reviews.

Outcomes
How will you address the issue of ROSC being recorded at different time points in studies? (e.g. any ROSC, ROSC>20 minutes, ROSC at hospital handover). Where multiple time points are used, which will you use?
We clarified in the “assumptions and simplifications” section that we only consider the outcome

measure “ROSC at any time” as “ROSC”, as stated in the “outcomes” section. We will not consider “sustained” ROSC for defined time intervals like “more than 20 minutes”: While this arbitary chosen interval may have influence on the patients’ further treatment options, we are convinced that the underlying cause for cardiac arrest will have a much greater impact on the duration of ROSC than the chest compression technique used to achieve ROSC. Therefore, we will not consider ROSC at time points different to those stated in our “outcomes” section. In the studies screened for our previous works, any study reporting ROSC at arbitrary chosen time points (e.g. after / longer than for example 20 minutes, 1 hour, or 4 hours, respectively) also reports ROSC “at any time”.

The outcome “ROSC at hospital handover” you mentioned is identical to our secondary outcome “alive at hospital admission” in the category “short-term survival”, though assessed separately.

Why is neurological outcome not considered after discharge/ 30-days?
Neurological “long-term” outcome is the outcome which we assume to be most relevant to our patients, but we identified only four studies reporting survival rates later than 30 days after the incident in a non-comprehensive preliminary analysis of this review: 2 studies reported 3-month, 1 study reported 6- and another 12-month survival rates (and not each of them does even report neurological outcome). As this endpoint is virtually never reported, we did not consider it for our review.

Please review your outcome list and check consistency with PROSPERO registration.

As you correctly point out, we did not update the PROSPERO record before submitting our manuscript, as we wanted to await if protocol amendments are demanded during the review process. Meanwhile, we submitted our updated record. Subsequently, the CRD-Register informed us that currently processing might take up to three months for submissions from outside the UK.

Eligibility criteria
The introduction notes the publication of three large randomised controlled trials. What is the rationale for including quasi-RCTs?
The authors are aware that quasi-RCTs are at risk of biases in random sequence generation,

blinding, and allocation concealment. Though, we decided to include quasi-RCTs because besides the fact that random sequence generation is not achieved, some of the other limitations of quasi-RCTs are not applicable here: Blinding of the rescues is impossible even in RCTs due to the intervention, and blinding of the population is achieved as the patients are concoiusless. Though, allocation concealment is rather difficult to assign with this intervention.

On the other hand, most Cochrane Reviews only include RCTs, and their evidence is already known. Yet, there are several smaller qasi-RCTs which might be considered, and we would then be able to compare our results to those from existing “RCT-only” Reviews. Especially Emergency Medicine is a setting which makes it difficult plan trials depicting clinical practice and reality. In these cases, including quasi-RCTs may provide an additional and valuable input from “real world evidence”, though contributing to a lager overall picture.

General comments

It would be informative for the reader to provide more information about the timeline of this project.

We detailed our scheduled timeline in the “Ethics and Dissemination” section. We state the expected duration of the review in months only (without setting any exact date), as we expect current COVID-19 development with dramatically increasing case numbers in Germany will commit the reviewers and co-authors to their clinical duties.

The PROSPERO record is recorded as being completed in 2017, with a start date of 2016.
The record also states: “An abstract, and oral presentation containing partial aspects of a preliminary analysis have been publicized in German language at the scientific meeting “Deutscher Anästhesie Congress” in Düsseldorf, Germany, on May 7th, 2015.”
Our preliminary analysis was executed in the context of the corresponding authors’ doctoral thesis mentioned in the “Ethics and dissemination” section and presented at a scientific meeting. We were not able to meet the scheduled timetable as our co-authors from the department of Anaesthesiology had been asked to get involved in the organization of the departments’ 50th anniversary, and in writing a chronical for this occasion. Due to clinical obligations, we unfortunately had been forced to postpone this project. But as we still are convinced that – despite several high-quality reviews have been added

the last years – this rewiew might contribute to close the knowledge gap and to put our focus from the rather general question “which technique is superior?” to the individualized question “which patients under which conditions profit from either one or the other technique?”.

Search strategy

It would be helpful to provide information on how you developed your search strategy. Was an information scientist involved? Did you consider the inclusion of MESH terms (e.g. “cardiopulmonary resuscitation” with the instrumentation qualifier)?

Our search strategy is based on the search strategy from our pilot analysis mentioned above. Two information scientists were involved in the advancement of the search strategy for the purpose of this systematic review and meta-analysis. We did not explicitely state their involvement in the methology section, as they are co-authors of this manuscript: Ms. Seide is a biometrician with expertise in meta-anaysis, and Dr. Jensen is head of the meta-analysis workgroup at the Heidelberg University Institute of Medical Biometry and Informatics.

MeSH terms (“resuscitation”, “resuscitator”) are indicated with a slash (“/”) in OvidSP, we now point this out in the legend to table 4.

Public and patient involvement

It would be more appropriate to simply detail that patients and the public were not involved. Many studies of cardiac arrest have successfully included patients and the public in their design and oversight (e.g. PARAMEDIC trial).

We deleted the first to paragraphs of this section and took over the statement as proposed.

General readability

I would recommend careful proof-reading of the paper.

The co-authors again proof-read the manuscript carefully and corrected mistakes.

VERSION 2 – REVIEW

REVIEWER	R. Brandon Hunter, MD and Ryan W. Morgan, MD, MTR Children's Hospital of Philadelphia, United States of America
REVIEW RETURNED	24-Nov-2020

GENERAL COMMENTS	Summary of Improvements: I appreciate the authors' response to my initial suggestions. They have better highlighted the subgroup analyses that they intend to perform to identify subpopulations that may benefit from mechanical chest compressions. They also more explicitly explained how their study will build upon and differentiate itself from prior meta-analyses. It is still unclear whether the addition of quasi-RCT data to the recent Cochrane review will allow for successful execution of these subgroup analyses but it is a worthy endeavor. The team has done an excellent job condensing the paper, especially the earlier sections. I think this revision is significantly improved. I would ensure that the objectives of the paper are clearly and consistently stated throughout. Specific Comments and suggestions: - Abstract line 47 – since the subgroup analysis is what sets this review apart from prior studies per the response to reviewers, would highlight it by mention in the abstract - Lines 99-104 – crisp and improved introduction - Lines 106-108 – thank you for condensing this section - would consider adding a single follow-up sentence to the effect of "mechanical CPR may represent a method of improving compression quality in certain clinical scenarios and for chose subpopulations" to complete the rationale for the study - Line 162 – I think it should be "and over manual CPR" (rather than mechanical) though authors should confirm - Line 236 – The statement "…and the quantity of viable cells within the organism" is a bit of a stretch regarding the value of ETCO2 - would just refer to its relationship to cardiac output and leave this phrase out unless there is a specific citation to this effect. - Line 192, Objectives – I think this is a good point to be as clear as possible with review questions and objectives – if urban vs rural, US vs Europe vs other, in-hospital vs out-of-hospital are the subgroups to be analyzed, should be listed explicitly (could do so under Patients or Review Questions)

REVIEWER	Keith Couper University of Warwick
REVIEW RETURNED	13-Nov-2020

GENERAL COMMENTS	Thank you for your efforts in addressing the first round of reviewer comments. I have a few outstanding issues: - The rationale (lines 105-108) is true, but seems to ignore middle-aged/ older people and disabilities that may also have the potential for recovery with good quality of life. - The choice of outcomes is obviously a decision for the authors. However, the argument that an outcome should not be included because it is infrequently reported seems flawed. A key finding of any review is to look for knowledge gaps. The core outcome set for cardiac arrest (COSCA) describes the outcomes that should be reported in all cardiac arrest clinical trials. If the review finds that

	these outcomes are not reported in mechanical CPR trials, this is an important finding. - It states in your outcome section that “Neurological outcome is evaluated after 30 days, or at hospital discharge, whichever comes first.” Most trials will typically select a single time point rather than use this definition. - It seems an unusual approach to exclude studies that report survival outcomes, but do not report neurological outcome or ROSC (see lines 262-263). - I am unclear what you mean by “Any prospective study may be subject to the Hawthorne Effect, when participants show a different behaviour due to the knowledge that they participate in a study”- how is this relevant in the context of a study where the participant is in cardiac arrest? Are you referring to performance bias? - There continue to be spelling and grammatical errors.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1

Summary of Improvements: I appreciate the authors’ response to my initial suggestions. They have better highlighted the subgroup analyses that they intend to perform to identify subpopulations that may benefit from mechanical chest compressions. They also more explicitly explained how their study will build upon and differentiate itself from prior meta-analyses. It is still unclear whether the addition of quasi-RCT data to the recent Cochrane review will allow for successful execution of these subgroup analyses but it is a worthy endeavor. The team has done an excellent job condensing the paper, especially the earlier sections. I think this revision is significantly improved. I would ensure that the objectives of the paper are clearly and consistently stated throughout.

We are very grateful for your encouraging feedback.

Specific Comments and suggestions:

- Abstract line 47 – since the subgroup analysis is what sets this review apart from prior studies per the response to reviewers, would highlight it by mention in the abstract

We now mention our subgroup analysis already in the abstract.

- Lines 99-104 – crisp and improved introduction

- Lines 106-108 – thank you for condensing this section - would consider adding a single follow-up sentence to the effect of “mechanical CPR may represent a method of improving compression quality in certain clinical scenarios and for chose subpopulations” to complete the rationale for the study

Thank you very much for your suggestion. We added this between “introduction” and “rationale”.

- Line 162 – I think it should be “and over manual CPR” (rather than mechanical) though authors should confirm

Of course, it should read “manual”.

- Line 236 – The statement “…and the quantity of viable cells within the organism” is a bit of a stretch regarding the value of ETCO2 - would just refer to its relationship to cardiac output and leave this phrase out unless there is a specific citation to this effect.

As suggested, we let this part out.

- Line 192, Objectives – I think this is a good point to be as clear as possible with review questions and objectives – if urban vs rural, US vs Europe vs other, in-hospital vs out-of-hospital are the subgroups to be analyzed, should be listed explicitly (could do so under Patients or Review Questions)

We detailed any intended subgroup analysis in the “Data synthesis” section (as suggested in item #15c of the PRISMA-P checklist). To avoid double-mentioning, we added the characteristics to the “Objectives” section.

Reviewer: 2

Thank you for your efforts in addressing the first round of reviewer comments.

We appreciate your suggestions which contributed to make our study protocol more accurate and appropriate.

I have a few outstanding issues:

- The rationale (lines 105-108) is true, but seems to ignore middle-aged/ older people and disabilities that may also have the potential for recovery with good quality of life.

We became aware that our wording (also in the “burden of disease” section) focussed rather on the economic than on the social burden, and clearified our message in both sections.

- The choice of outcomes is obviously a decision for the authors. However, the argument that an outcome should not be included because it is infrequently reported seems flawed. A key finding of any review is to look for knowledge gaps. The core outcome set for cardiac arrest (COSCA) describes the outcomes that should be reported in all cardiac arrest clinical trials. If the review finds that these outcomes are not reported in mechanical CPR trials, this is an important finding.

We understand your criticisms and it seems we might have taken a too pragmatic approach by limiting the outcomes to those reported in previous studies. Of course, the suggested procedure would be the transparent and proper approach, so we followed your suggestions and added the neurological / quality of life outcomes to the pure long-term survival rate.

- It states in your outcome section that “Neurological outcome is evaluated after 30 days, or at hospital discharge, whichever comes first.” Most trials will typically select a single time point rather than use this definition.

Trials usually report one of both time points, but do not consistently use the same definitions. Each definition is acceptable and has its own advantages and drawbacks - and both are intended to map the outcome once treatment of the underlying cause is introduced or completed (simplified, patients are discharged, dead, or suffered complications leading to long-term course). By deleting our addition “whichever comes first” (which we initially added in order to define what we intend to do if outocmes for both defintions were reported) we reflect the Utstein/COSCA defintions more accurately.

- It seems an unusual approach to exclude studies that report survival outcomes, but do not report neurological outcome or ROSC (see lines 262-263).

We totally agree with your reservations. Our definition is based on the assumption that any resuscitation study reporting short- or long-term survival outcomes will also report ROSC as the

easiest-to-obtain “survival” outcome and therefore meet inclusion criteria. We clarified this by specifically mentioning survival outcomes.

- I am unclear what you mean by “Any prospective study may be subject to the Hawthorne Effect, when participants show a different behaviour due to the knowledge that they participate in a study”- how is this relevant in the context of a study where the participant is in cardiac arrest? Are you referring to performance bias?

By this sentence we refer to the effect when “individuals” (in this sense of course not the participating patients, but the participating emergency care teams) modifiy their behavior when they are aware of being observed. While “performance bias” may suggest better/worse treatment in one specific group, this effect may also occur in both, intervention and control groups, likewise.

- There continue to be spelling and grammatical errors.

We had our manuscript critically proof-read again. Several sentences have been put more clearly and typos have been corrected.

VERSION 3 – REVIEW

REVIEWER	R. Brandon Hunter, MD and Ryan W. Morgan, MD, MTR Children's Hospital of Philadelphia Perelman School of Medicine at the University of Pennsylvania Philadelphia, Pennsylvania, United States
REVIEW RETURNED	05-Jan-2021

GENERAL COMMENTS	My comments have been addressed. The manuscript's grammar and flow is globally improved as well.

REVIEWER	Keith Couper University of Warwick, Coventry, UK
REVIEW RETURNED	05-Jan-2021

GENERAL COMMENTS	Thank you- I believe my comments have now been addressed,